Wall bracket (cantilever)

Plane truss — analysis and CIRSOC 301 rod checks

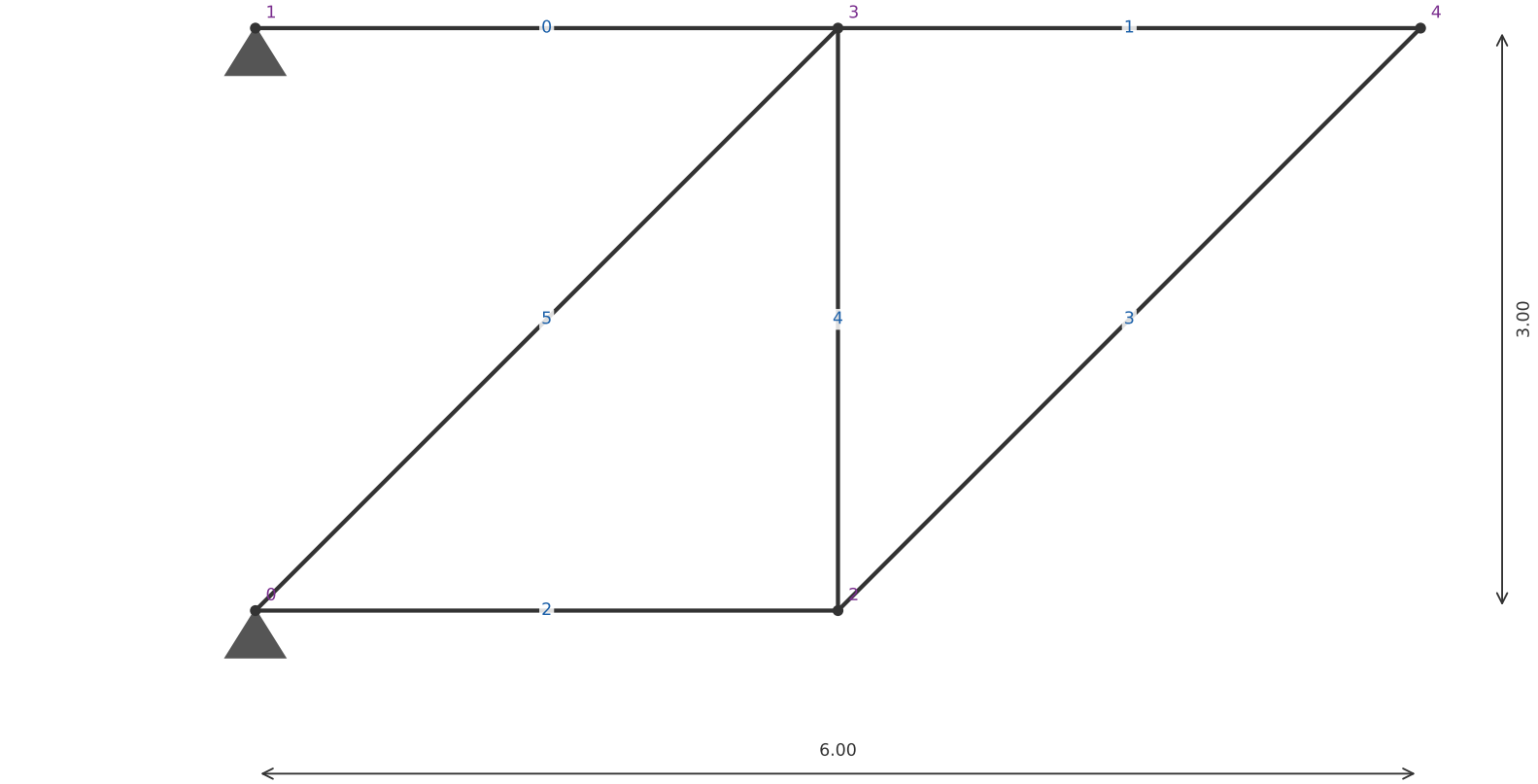
Span	6.00 m
Height	3.00 m
Nodes / rods	5 / 6
Supports	pin at 0, pin at 1
Loads	1 nodal, total 25.0 kN
Family "Chord"	CHS 101.6x4.0 (F24 (CIRSOC) / A36)
Family "Web"	CHS 76.1x3.6 (F24 (CIRSOC) / A36)
Weight	Weighs 178 kg, carries 25.0 kN (14 × its own weight), passes (0.76).
Deflection	4.45 mm = L/1348, limit L/300: OK
Stability	Statically determinate (m + r − 2j = 6 + 4 − 2×5 = 0): statics alone gives every force.
Most used rod	rod 5 at 0.76 (elastic buckling (E3-3))
Rods over capacity	0 of 6 checked
Buckling	$\lambda_1 = 1.69$
Shear and moment	none: every rod is pinned and loaded only at its nodes, so the axial force is the whole result

Results are an aid to a qualified professional, who must check them. Rod checks to CIRSOC 301 as implemented; first-order analysis; joints and connections are not checked here.

Contents

2. Geometry
3. Axial forces
4. Deflection and stress
5. Reactions
6. Utilisation
7. Buckling
8. Rods
9. Explain rod 5 and 3

Node numbers in purple, rod numbers in blue. Dimensions in m.

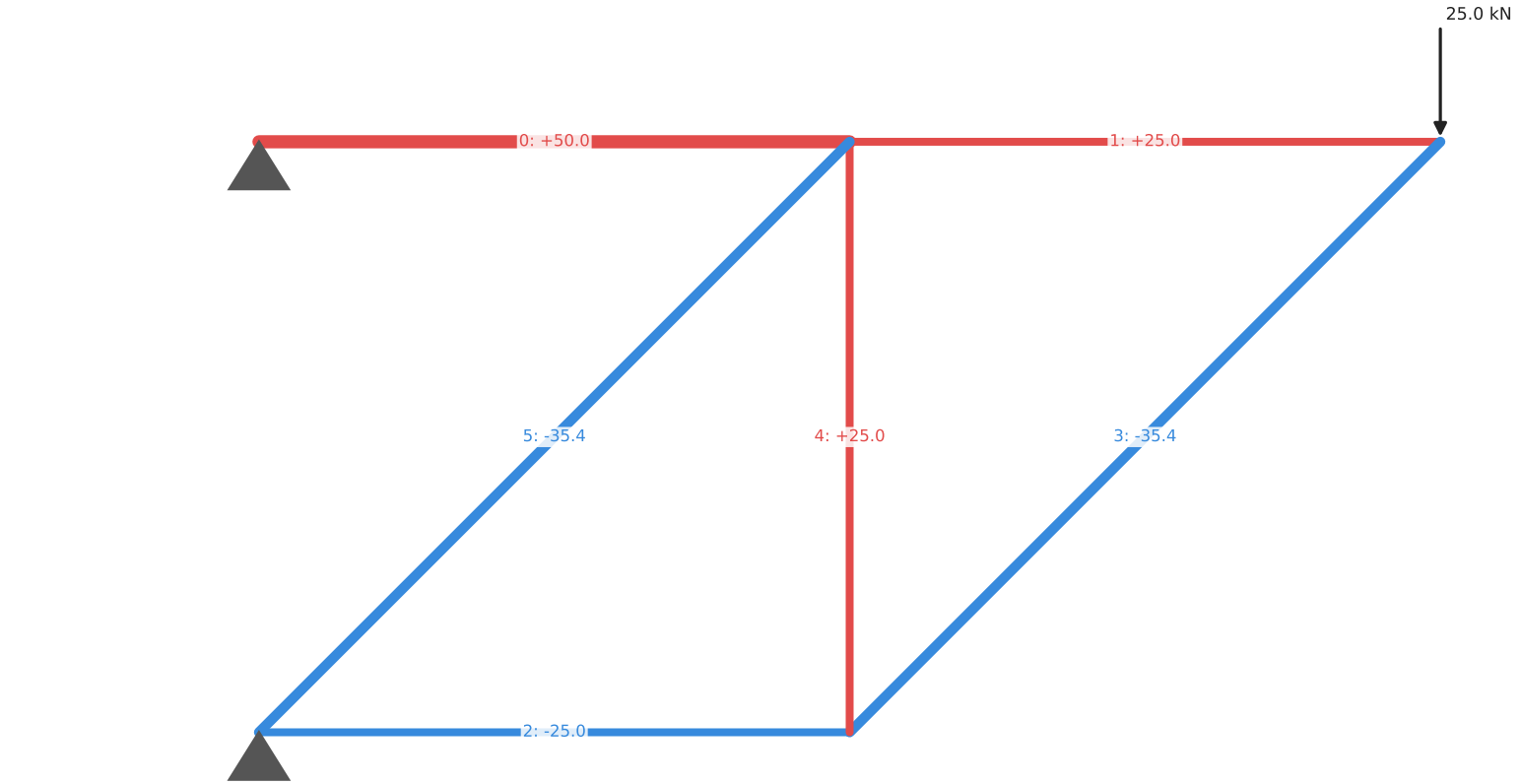


Truss — Geometry

Sheet 2 / 9

MODEL	SIZE	UNITS	CODE	DATE
Wall bracket (cantilev	5 nodes / 6 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

Axial force N (kN): red pulls (+), blue pushes (−), grey carries nothing; line width follows |N| (largest 50.0).



2 m · true to scale

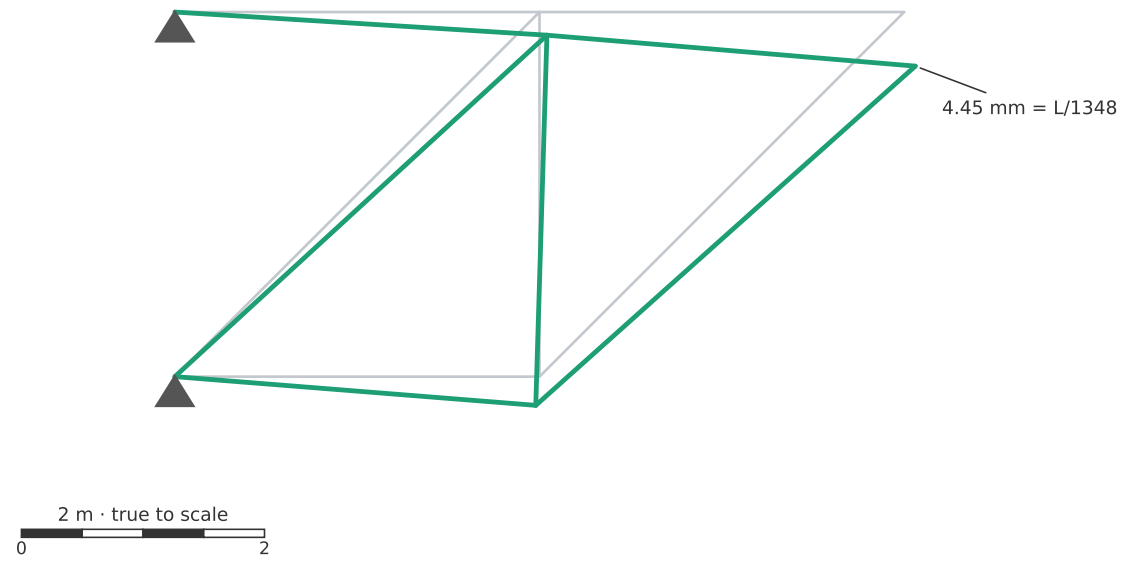
0 2

Truss — Axial forces

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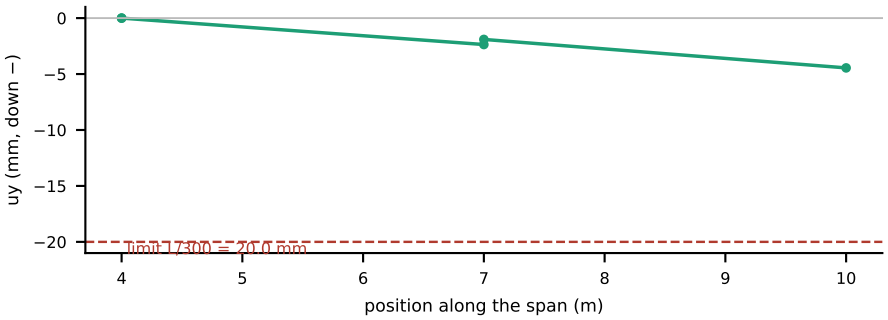
MODEL	SIZE	UNITS	CODE	DATE
Wall bracket (cantilev	5 nodes / 6 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

Deflected shape (green) over the truss (grey). Geometry to scale; displacements magnified × 100. Largest vertical deflection 4.45 mm against the limit L/300 = 20.0 mm: OK.



Elastic deformation

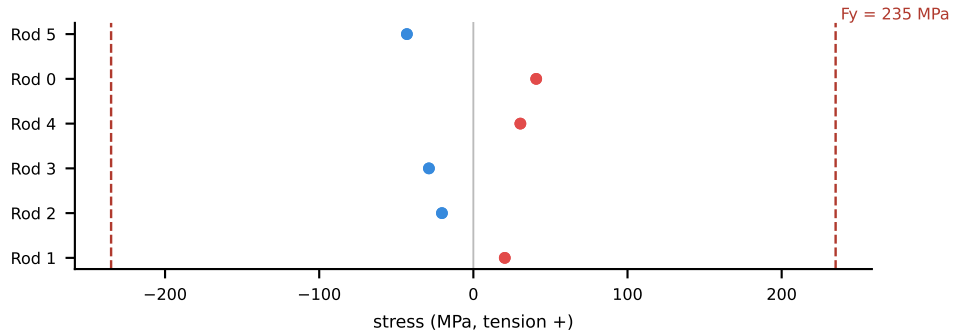
vertical movement of every joint



Largest 4.45 mm at joint 4 = L/1348. Within the limit.

Fibre stress

N/A ± M·c/I at each rod's extreme fibres



Every rod is pinned, so its two fibres carry the same N/A and coincide: no bending opens the pair.

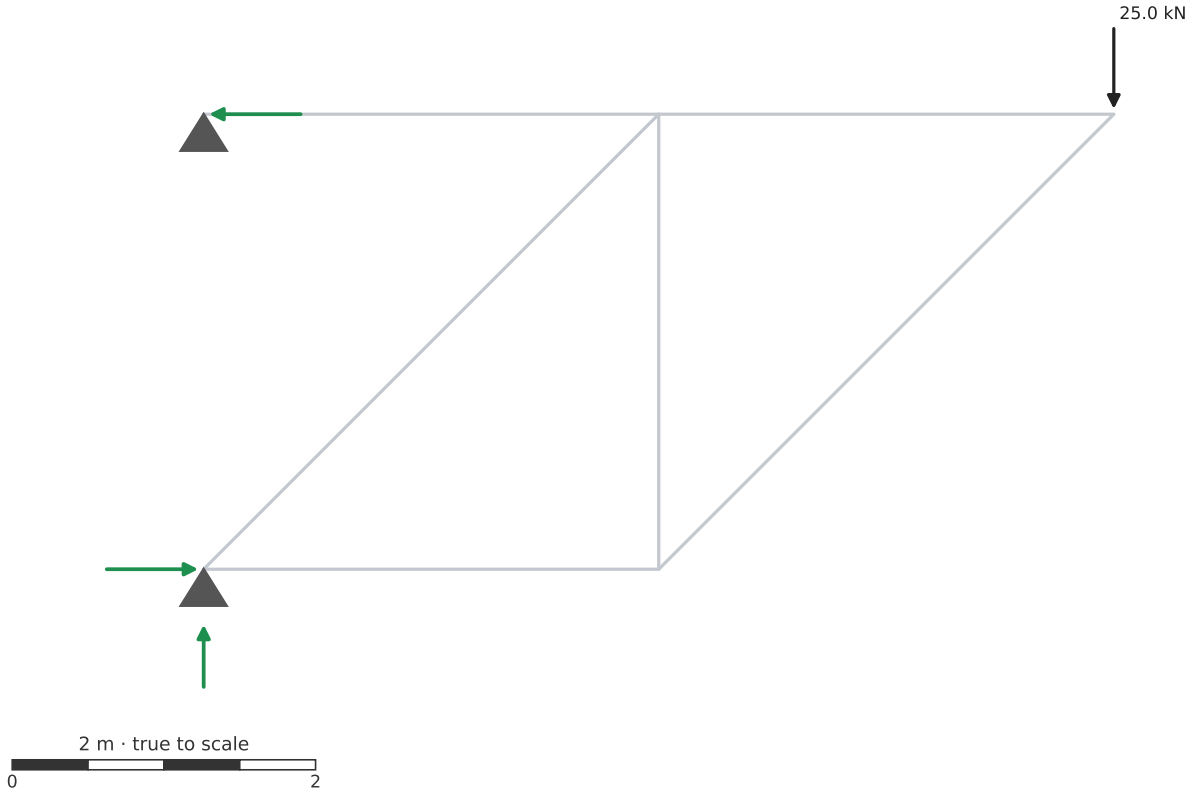
Truss — Deflection and stress

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MODEL	SIZE	UNITS	CODE	DATE
Wall bracket (cantilev	5 nodes / 6 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

Support reactions (green arrows), as the supports push on the truss; Ry positive upwards. The sums balance the loads.

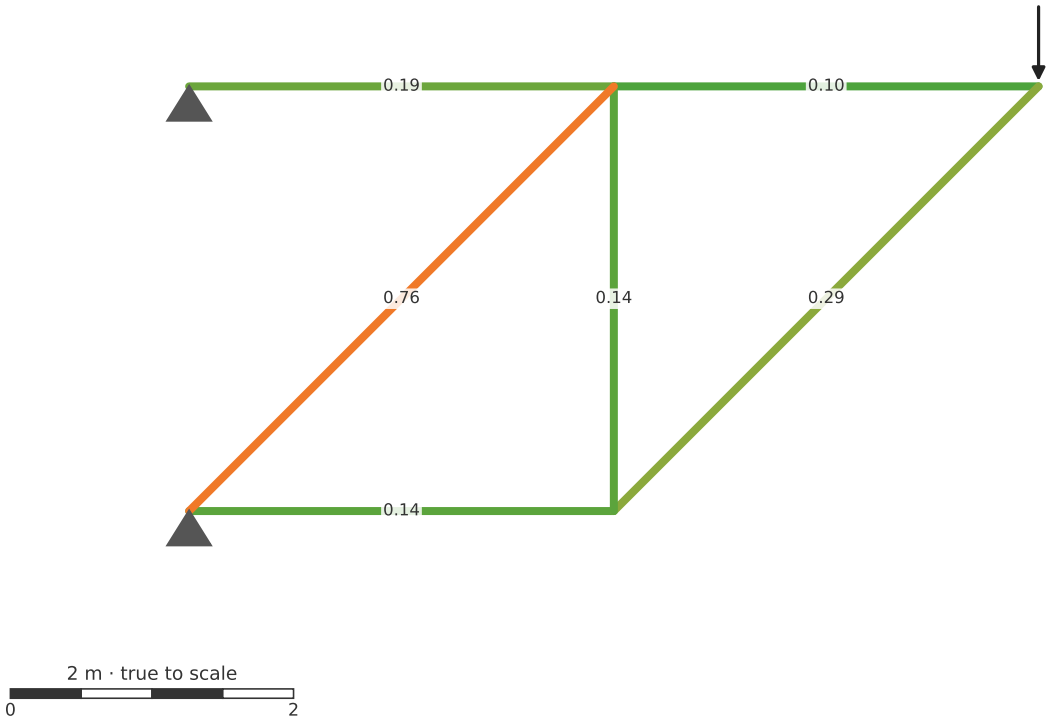
Node	Type	Rx (kN)	Ry (kN)
0	pin	50.00	25.00
1	pin	-50.00	0.00
Σ		0.00	25.00

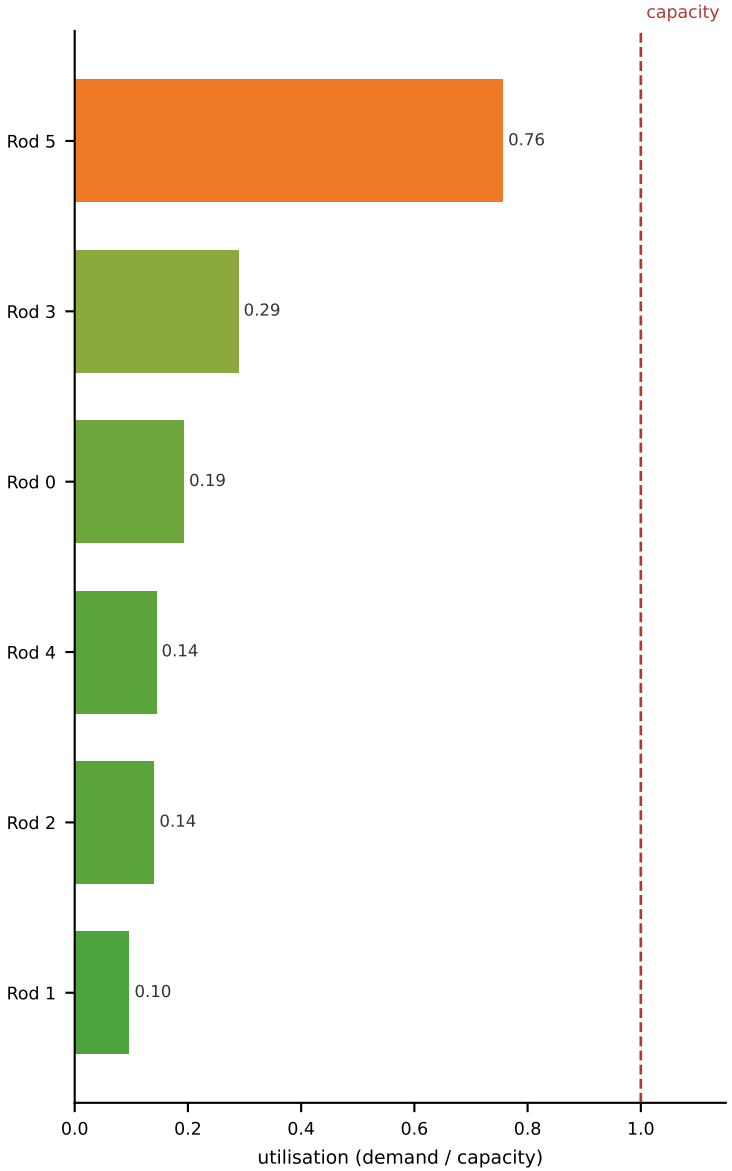


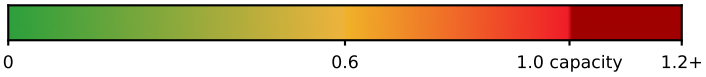
Truss — Reactions

MODEL	SIZE	UNITS	CODE	DATE
Wall bracket (cantilev	5 nodes / 6 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

Utilisation: demand over CIRSOC 301 capacity. Most used rod 5 at 0.76 (elastic buckling (E3-3)). Every checked rod is within capacity.







grey = no steel section, not checked

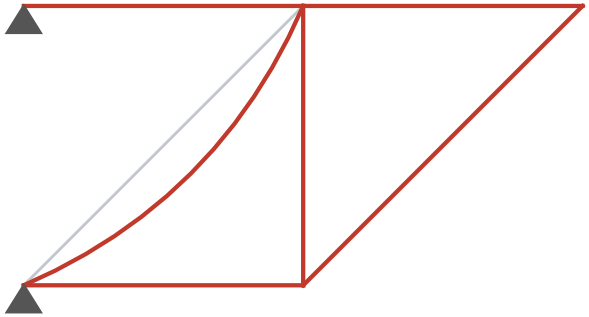
Truss — Utilisation

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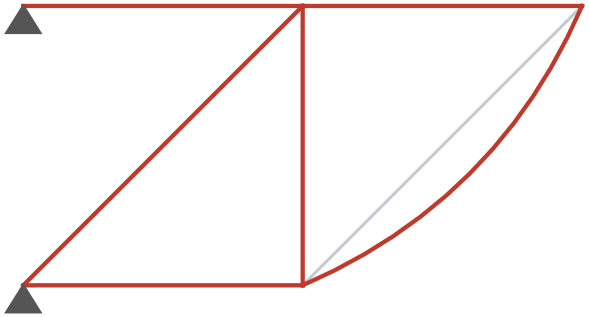
MODEL	SIZE	UNITS	CODE	DATE
Wall bracket (cantilev	5 nodes / 6 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

λ is the factor on the load as it stands. $\lambda_1 = 1.69$, so the load can grow 1.7× before it buckles. Mode 2 is 4.57, 171 % higher: one member decides the buckling load.

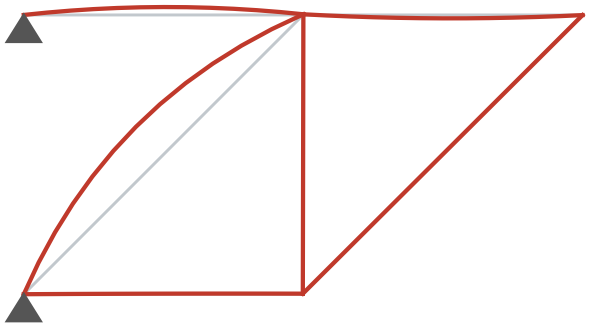
Mode 1: λ = 1.69
rod 5 bows

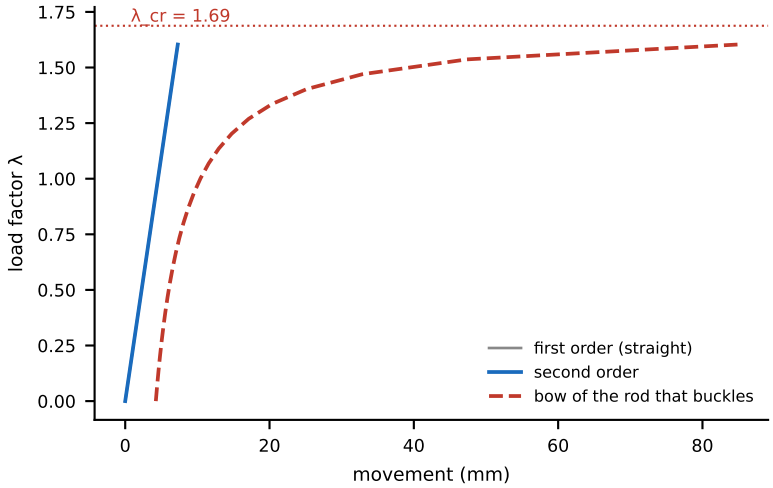


Mode 2: λ = 4.57
rod 3 bows



Mode 3: λ = 8.15
rod 5 bows





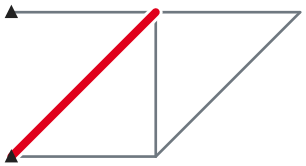
The second-order line bends away from the first-order one and runs off as λ nears λ_cr: that gap IS the amplification every second-order design rule is about. Started from an initial bow of L/1000 = 4.24 mm in the first mode's shape.

Elastic buckling of the IDEAL truss, so every λ is an UPPER bound: imperfections, residual stresses and yielding are not modelled, and each rod is cut in two so it can bow between its joints. Buckling OUT of the plane is the rod check's job (CIRSOC 301 E3, minor radius) and is in the Utilisation and Rods sheets, not here. A mode has a shape, not a size: the shapes are drawn exaggerated.

Truss — Buckling

MODEL	SIZE	UNITS	CODE	DATE
Wall bracket (cantilev	5 nodes / 6 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

Rod	Nodes	Family	Section	L (m)	N (kN)	Governing check	Use	
0	1–3	Chord	CHS 101.6x4.0	3.00	+50.00	normal stress (H.3.4)	0.19	OK
1	3–4	Chord	CHS 101.6x4.0	3.00	+25.00	normal stress (H.3.4)	0.10	OK
2	0–2	Chord	CHS 101.6x4.0	3.00	-25.00	inelastic buckling (E3-2)	0.14	OK
3	2–4	Chord	CHS 101.6x4.0	4.24	-35.36	inelastic buckling (E3-2)	0.29	OK
4	2–3	Web	CHS 76.1x3.6	3.00	+25.00	normal stress (H.3.4)	0.14	OK
5	0–3	Web	CHS 76.1x3.6	4.24	-35.36	elastic buckling (E3-3)	0.76	OK



Rod 5

utilisation 0.76

elastic buckling (E3-3)

```
Rod 5
=====
Steel, CIRSOC 301-2018 -- A = 820 mm², Fy = 235 MPa, L = 4.243 m

 1. Slenderness  [E2]
      K·L / r
      = 1 × 4,243 mm / 25.7 mm
      = 165.3

 2. Euler stress  [E3-4]
      Fe = π² E / (KL/r)²
      = π² × 200,000 / 165.3²
      = 72.23 MPa

 3. Which branch  [E3]
      KL/r ≤ 4.71 √(E/Fy) ?
      = 165.3 vs 4.71 √(200,000/235) = 137.4
      = elastic

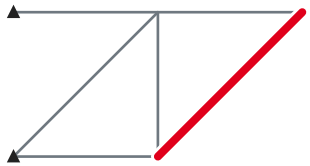
 4. Critical stress  [E3-3]
      Fcr = 0.877 · Fe
      = 0.877 × 72.23
      = 63.35 MPa

 5. Design strength  [E3-1]
      Pd = φc · Fcr · A
      = 0.9 × 63.35 × 820
      = 46.75 kN

 6. Compression ratio  [E3]
      |N| / Pd
      = 35.36 / 46.75
      = 0.756

 7. Governs
      the largest ratio
      = elastic buckling (E3-3)
      = 0.756

utilisation 0.756 -- OK
```



Rod 3

utilisation 0.29

inelastic buckling (E3-2)

```
Rod 3
=====
Steel, CIRSOC 301-2018 -- A = 1,226 mm², Fy = 235 MPa, L = 4.243 m

 1. Slenderness  [E2]
      K·L / r
      = 1 × 4,243 mm / 34.5 mm
      = 122.8

 2. Euler stress  [E3-4]
      Fe = π² E / (KL/r)²
      = π² × 200,000 / 122.8²
      = 130.80 MPa

 3. Which branch  [E3]
      KL/r ≤ 4.71 √(E/Fy) ?
      = 122.8 vs 4.71 √(200,000/235) = 137.4
      = inelastic

 4. Critical stress  [E3-2]
      Fcr = 0.658^(Fy/Fe) · Fy
      = 0.658^(235/130.80) × 235
      = 110.78 MPa

 5. Design strength  [E3-1]
      Pd = φc · Fcr · A
      = 0.9 × 110.78 × 1,226
      = 122.29 kN

 6. Compression ratio  [E3]
      |N| / Pd
      = 35.36 / 122.29
      = 0.289

 7. Governs
      the largest ratio
      = inelastic buckling (E3-2)
      = 0.289

utilisation 0.289 -- OK
```